PGR111, tasks for week 03

Task 1 - concepts and terminology

Briefly explain the following terminology and database concepts:

- Foreign key
- Primary Key
- The null value (“NULL”)
- Auto increment (“AUTO_INCREMENT”)

Write down the answers and save them. Might come in handy later in the semester? :-)

Task 2 - SQL

For the first 11 tasks, create a new MySQL sample database ("create schema"). Call it **People** and set it as "default schema". (Task 12+ uses World, as before.)

Write SQL against the new **People** database (schema) that performs the following:

1. Create a table called `person`. The table should contain:
   `ID, int`, autogenerated.
   `Ssn, char(11)`, should not be `null` and should be unique.
   `Name, varchar(100)`, should have default value `'Unknown'`.
   `Email, varchar(100)`, should be unique.
   Use the column `ID` as Primary Key.

   Check that the table is created by refreshing the object browser on the left in the MySQL workbench, then clicking *People* and its tables.

2. Enter a row in the person table. Specify the following data:
   Ssn: `'12345678901'`
   Name: `'Ola Nordmann'`
   Email: `'ola@nordmann.no'`

   Write an appropriate query to see that the data is in the table.

3. You decide that it's nice to record date of birth as well. Modify the existing `person` table, so that it also contains the column: `DateOfBirth`

   Check that the table has been updated to include the changes. (Same procedure as in task 1, or just write a query that displays all columns.)

4. Add a new row in the `person` table. Specify the following data:
   `Ssn: '98765432109'`

   Write an appropriate query to see that data resides in the table as expected.

5. Update the row with the ssn `'98765432109'`, so that the date of birth for the selected person will be: `'1990-01-01'`. (**Note:** Date fields are stored in MySQL with year first, then month, then day, as shown here.)

   Write an appropriate query to see that data is updated as expected.

6. Add yourself in the person table! Fill in all fields (you don't need to use real info if you don't want to).

7. We realize that our database is a bit to restricted in its design: Each person can only have 1 email address. Create a separate table called `email` with the following criteria:
   `Address, varchar(75)`, should not be `null` and should be unique.
   `Type, varchar(50)`. *(Example content: 'work', 'school', 'private', etc.)*
   `Person_ID, int`, should not be able to be `null`.
   For the primary key, use the `Address` column.
   As a foreign key, use `Person_ID` that reference the `person` table's column `ID`.

   Check that the table has been updated to include the changes. (Same procedure as in task 1, or just write a query that displays all columns.)

8. Write an SQL query that copies email addresses and IDs from the Person table to the Address table and to the `Person_ID` column in the Email table. *(**Tip:** You need to feed the insert with a query. Replace "values (...)" with "select...". If you're not sure how to do it, look at chapter 3.3.1 in the Norwegian textbook, or look it up online.)*

9. (If you have done task 8.)
   Now that we have moved the email addresses to a separate table, we can delete the email column from the `person` table. *Note*: We want to remove the entire column, not just the data in it.

10. Create a table like the one in task 1 but call it `personCopy`. (No need to copy data into it.) Check that the table is created. Then write the SQL command to delete it. *Note:* The idea is to test the command to delete a table, without destroying the two tables you've been working on for a while now. :-)

11. **DIFFICULT:** If we extract the name and date of birth from the person table, the date comes on MySQL's for us "backwards format". In a SELECT, reformat the output so date is shown on the format: 31-01-2001. (<-- Example of date with the correct format.)
    ***Tip:** Read about the MySQL function: DATE_FORMAT() (w3schools or another online guide)*

Add the date for Ola Nordmann. He should have DateOfBirth 20.02.1912 (old man!). Write it on this format ('dd.mm.yyyy'), use STR_TO_DATE() for MySQL to understand the format.

Check that the table has been deleted.

Use the MySQL sample database ("schema") **world** for the remaining tasks.
(Set **world** as "default schema".)

Write SQL for the **world** database (schema) that performs the following:

12 Delete the row containing the city 'Bærum' from the city table. (This is no city and should not be in the city table!).

Write an appropriate query to see that data is deleted as expected.

13 Use the max() function to retrieve the largest ID in world.city.

SELECT * from information_schema.tables for the world database (WHERE table_schema = 'world'). Can you find out how many rows you have in your different tables? What is the name of the column indicating this? Can you find the column that gives us the next value for city.ID?

Tip: Besides asking the teacher assistants, lecturer and using the forum on Canvas, look at http://www.w3schools.com/sql/default.asp. Here you will find answers to many of your SQL questions :-) Also remember that the textbook has many nice resources, including answers to questions from the book. You will learn a lot from solving the puzzles in the book too!

You can also spend time exploring other databases from the textbook, or in Workbench/online. (Ex: the SAKILA database.)